

BOLD - behaviour correlations

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data from BOLD amplitude Model 1

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/ROI/Hippocampus")
BOLD <- read.csv("BOLD_Hipp_model1.csv")

# Assign factors
BOLD$$ID <- factor(BOLD$ID)
BOLD$ID <- factor(BOLD$ID)
BOLD$ROI <- factor(BOLD$ROI)
BOLD$hemisphere <- factor(BOLD$hemisphere)
BOLD$StimType <- factor(BOLD$StimType)
BOLD$TaskStage <- factor(BOLD$TaskStage)

# select variables/conditions of interest
BOLD_m1 <- subset(BOLD, TaskStage=="Encode" & hemisphere=="L")
BOLD_m1 <- subset(BOLD_m1 , select = -c(hemisphere))
```

Import data from BOLD amplitude Model 2

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/ROI/Hippocampus")
BOLD <- read.csv("BOLD_Hipp_model2.csv")

# Assign factors
BOLD$$ID <- factor(BOLD$ID)
BOLD$ID <- factor(BOLD$ID)
BOLD$ROI <- factor(BOLD$ROI)
BOLD$hemisphere <- factor(BOLD$hemisphere)
BOLD$StimType <- factor(BOLD$StimType)
BOLD$TaskStage <- factor(BOLD$TaskStage)
BOLD$Response <- factor(BOLD$Response)

# select variables/conditions of interest
```

```
BOLD_m2 <- subset(BOLD, TaskStage=="Encode" & hemisphere=="L" & Response=="C>I")
BOLD_m2 <- subset(BOLD_m2 , select = -c(hemisphere))
```

Import data from Behaviour during fMRI

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/Behaviour")
load("Avg_Behv.Rda")

bev <- bev %>%
  rename(
    ID = S_ID)

bev_acc <- subset(bev, RespType=="Target")
```

Perform Correlations

```
# Combine dataframes
library(data.table)
B1_beh <- merge(BOLD_m1, bev_acc, by=c("ID", "StimType"))

# Correlations - between BOLD signal model 1 and behavioural performance
corr1 <- ddp1y(B1_beh, .(StimType, ROI, RespType), corrgroup, "BOLD", "mean")
pander(corr1)
```

StimType	ROI	RespType	COR	p.value
Sham	Ant	Target	-0.07085	0.7666
Sham	Hippocampus	Target	-0.01305	0.9565
Sham	Mid	Target	-0.2888	0.2168
Sham	Post	Target	-0.07891	0.7409
TI 1:1	Ant	Target	0.07367	0.7576
TI 1:1	Hippocampus	Target	0.1393	0.5582
TI 1:1	Mid	Target	0.01153	0.9615
TI 1:1	Post	Target	0.1273	0.5928
TI 1:3	Ant	Target	0.1588	0.5036
TI 1:3	Hippocampus	Target	0.22	0.3512
TI 1:3	Mid	Target	0.02905	0.9033
TI 1:3	Post	Target	0.2996	0.1994

```
# no dignificant correlations
```

```
# -----
# Correlations - between BOLD signal model 2 and behavioural performance

B2_beh <- merge(BOLD_m2, bev_acc, by=c("ID", "StimType"))
corr2 <- ddp1y(B2_beh, .(StimType, ROI, RespType), corrgroup, "BOLD", "mean")
pander(corr2)
```

StimType	ROI	RespType	COR	p.value
Sham	Ant	Target	0.176	0.458
Sham	Hippocampus	Target	0.3292	0.1564
Sham	Mid	Target	0.3223	0.1658
Sham	Post	Target	0.1662	0.4837
TI 1:1	Ant	Target	0.005973	0.9801
TI 1:1	Hippocampus	Target	-0.0429	0.8575
TI 1:1	Mid	Target	-0.03798	0.8737
TI 1:1	Post	Target	-0.01115	0.9628
TI 1:3	Ant	Target	0.2182	0.3554
TI 1:3	Hippocampus	Target	0.2247	0.3408
TI 1:3	Mid	Target	0.1531	0.5193
TI 1:3	Post	Target	0.1738	0.4636

no dignificant correlations